

Solution Requirements

Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Team ID	SWTID-2026-1254
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Functional Requirements

1. Accept story prompt and personalization inputs. 2. Validate form/JSON input. 3. Generate a structured 5-panel outline. 4. Generate detailed narration and character dialogue. 5. Generate comic-style illustrations. 6. Build a panel layout. 7. Preview the comic. 8. Export the comic as PDF. 9. Provide API endpoints for supported generation/testing workflows. 10. Handle errors using FastAPI HTTPException and try-except blocks.

Input Requirements

Story Prompt, Character Name, Setting, Story Tone, and Art Style.

Output Requirements

Panel-by-panel comic content, illustrations, preview page, and downloadable PDF.

Non-Functional Requirements

User-friendly interface, modular architecture, maintainable Python code, dynamic template rendering, local deployment support, and protected API credentials through environment variables.